

Blender Smoothie Cycle SOP (1x Batch: 2 Cups)

Prepare one smoothie batch per episode. Open the button-release lid, load the **fruit**, add the **milk**, push the lid closed, and press the **BLENDING BUTTON** for 30 seconds. Add 10 seconds if visible fruit pieces remain. Open the lid, set the pitcher on the table, and fill both serving cups while they remain in their right-side positions. Fill each cup to 1–2 cm below its rim. Set the pitcher back on the table after pouring; do not relock it.

Never place a gripper or object inside the pitcher or touch it while the blades move.

The table is set up in one of three ways. Only the ingredient cups move; the blender, the pitcher set-down spot, the serving cups, and the spill-cleanup spot are in the same place in all three.

- **Config L:** the ingredient cups are at the left, along the left table edge.
- **Config M:** the ingredient cups are at the front-center, between the pitcher set-down spot and the front table edge.
- **Config R:** the ingredient cups are at the back-right, right of the blender base.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the ingredient cups start at the left (**Config L**), the front-center (**Config M**) or the back-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the ingredient cups' side loads them — the **left gripper** in Config L and M, the **right gripper** in Config R. The other gripper steadies the pitcher: by the handle in Config L and M, by the lower wall opposite the handle in Config R. No arm reaches across the table.
- **Fixed roles:** everything else is the same in all three configs — the right gripper presses **LID RELEASE** and the **BLENDING BUTTON**, the left gripper pushes the lid closed, both grippers carry and pour the pitcher with the right gripper on the handle and the left on the lower wall, and the right gripper takes the cloth.
- **Order:** fruit first, milk second; the left serving cup is filled before the right one.

Setup

Complete both checklists before starting the episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows both table edges, the input zone, blender and buttons, pitcher set-down spot, two right-side serving cups, and spill-cleanup spot.
3. Both arms are at the home position with grippers open.
4. The blender is plugged in, off, and dry. The empty pitcher is seated on the base with its hinged lid pushed closed.
5. The table is clean, dry, and clear except for the task materials.

Materials checklist

1. One pre-portioned set is staged left to right in the input zone:
 - one open cup containing fruit; and
 - one open cup containing milk. Return each empty cup upright to its starting position. The input zone is one of:
 - **Config L:** the left, along the left table edge.
 - **Config M:** the front-center, between the pitcher set-down spot and the front table edge.
 - **Config R:** the back-right, right of the blender base.
2. Both cups are rigid and dry and empty through the pitcher opening without scraping.
3. The transparent pitcher begins empty and seated on the base with its handle facing right.
4. Its clean, dry hinged lid has a visible **LID RELEASE** button, stays open by itself, and closes when pushed down.
5. The blender has one validated automatic speed and one visible **BLENDING BUTTON**.
6. Two identical, clean, dry serving cups begin upright in their right-side pouring positions.
7. One absorbent cloth and one waste tray are staged at the front spill-cleanup spot. Spilled ingredients or smoothie are never returned to the pitcher or cups.
8. Confirm the full ingredient set fits inside the pitcher and fills both serving cups.

Workspace layout

- **Input zone:** two ingredient cups in fixed loading order: fruit, milk — at the left (Config L), the front-center (Config M), or the back-right (Config R); return each empty cup to its original position
- **Center back:** blender base with the pitcher seated and lid pushed closed
- **Center front:** clear pitcher set-down spot

- **Right:** two serving cups in their pouring positions
- **Front-right:** spill-cleanup spot with cloth and waste tray

Vocabulary

- **Ingredient set:** one fruit cup and one milk cup.
- **Input zone:** where the ingredient set stands at the start of the episode — the left (**Config L**), the front-center (**Config M**), or the back-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Loading gripper:** the gripper on the ingredient cups' side — the left gripper in Config L and M, the right gripper in Config R. It lifts, tips, and returns each ingredient cup.
- **Fully emptied cup:** no loose fruit piece or pool of milk remains.
- **Open lid:** released by the **LID RELEASE** button and held fully open without support.
- **Closed lid:** pushed fully down against the pitcher.
- **Stopped blades:** no blade or smoothie motion is visible.
- **Blend complete:** with the blades stopped, no visible fruit piece remains.
- **Controlled pour:** the spout stays over the cup and the stream stays inside the rim.
- **Correct cup fill:** the smoothie surface is 1–2 cm below the cup rim.
- **Equal fill:** both cups are filled to the same visible level.
- **Abnormal blend condition:** raised lid, leak, shifting pitcher or base, or repeated hard-object strikes.

Steps

Load ingredients in this fixed order: **fruit first, milk second**.

Only Steps 2 and 3 depend on the config: the **loading gripper** takes each cup from the input zone (left in Config L and M, right in Config R) while the other gripper steadies the pitcher. Every other line is the same in all three configs.

Step 1: Open the lid

Goal: the hinged lid is fully open before loading begins.

- With the **right gripper**, press and release the **LID RELEASE** button.
- Move the right gripper clear while the lid opens.

Check: the lid stays fully open without support. If it does not open, press **LID RELEASE** once more with the **right gripper**. Do not load ingredients under a closed or partly open lid.

Step 2: Load the fruit

Goal: all fruit is in the empty pitcher and the cup is returned.

2.1 Stabilize the pitcher and lift the fruit cup

Look where the ingredient cups are before reaching for the fruit cup.

- **IF the ingredient cups are at the left (Config L):** with the **right gripper**, hold the pitcher handle. With the **left gripper**, grasp the fruit cup at its middle, lift it clear of the table, and center it over the pitcher.
- **IF the ingredient cups are at the front-center (Config M):** with the **right gripper**, hold the pitcher handle. With the **left gripper**, grasp the fruit cup at its middle, lift it clear of the table, carry it back past the set-down spot, and center it over the pitcher.
- **IF the ingredient cups are at the back-right (Config R):** with the **left gripper**, support the pitcher's lower wall opposite the handle. With the **right gripper**, grasp the fruit cup at its middle, lift it clear of the table, and center it over the pitcher.

2.2 Load the fruit and return the cup

Then, in all three:

- With the **loading gripper**, tilt the cup until all fruit falls into the pitcher. Do not shake it or strike the pitcher.
- With the **loading gripper**, return the cup upright to its starting position, then release the pitcher with the other gripper.

Check: the cup is empty, the rim is clear, and the pitcher is seated upright. With the **loading gripper**, tilt again for fruit left in the cup or guide a rim piece inward with the cup edge. Return the cup to its starting position. Never reach inside. Clean minor spills in Step 10.

Step 3: Add the milk

Goal: all milk is in the pitcher and both ingredient cups are returned.

3.1 Stabilize the pitcher and lift the milk cup

- **IF Config L or M:** with the **right gripper**, hold the pitcher handle and, with the **left gripper**, grasp the milk cup at its middle. **IF Config R:** with the **left gripper**, support the pitcher's lower wall opposite the handle and, with the **right gripper**, grasp the milk cup at its middle.
- Lift the cup clear of the table and center it over the pitcher.

3.2 Empty and return the milk cup

- With the **loading gripper**, tilt the cup slowly and pour until it is empty.
- Rotate the cup upright and return it with the **loading gripper**; then release the pitcher with the other gripper.

Check: both ingredient cups are empty, the rim is clear, and the pitcher remains seated. If milk remains, tilt the cup again with the **loading gripper**, then return it. Clean minor spills in Step 10.

Step 4: Push the lid closed

Goal: the hinged lid is fully closed before blending.

- With the **left gripper**, push the lid straight down until it rests fully closed, then release.
- Move both grippers clear of the pitcher.

Check: the lid rests fully closed against the pitcher. If any part is raised, push it down once more with the **left gripper**. Do not blend with an open or raised lid.

Step 5: Blend and extend once if needed

Goal: blend for 30 seconds, adding 10 seconds only if visible fruit pieces remain.

5.1 Start the blender

- Confirm both grippers are outside the pitcher's vertical footprint.
- With the **right gripper**, press and release the **BLENDING BUTTON**, then move clear.

5.2 Monitor and stop

- Run for 30 seconds. Do not touch the lid, pitcher, or base while the blades move.
- After 30 seconds, press and release the **BLENDING BUTTON** with the **right gripper**, move clear, and wait until no blade or smoothie motion is visible.

5.3 Inspect and run the 10-second extension if needed

- Inspect through the pitcher wall without touching the blender or lid.
- If no fruit piece is visible, continue to Step 6. Otherwise, move both grippers outside the pitcher's footprint.
- With the **right gripper**, press and release the **BLENDING BUTTON**, then move clear.
- Run for 10 seconds without touching the blender. Then press and release the **BLENDING BUTTON** with the **right gripper**, move clear, and wait until all motion stops.
- Inspect again. Never run a third cycle. If a fruit piece remains, do not open the lid or pour; end unsuccessful and inspect the ingredients and blender with recording off.

Abnormal-condition branch: immediately press and release the **BLENDING BUTTON** with the **right gripper**, clear both arms, and wait until all motion stops. Do not resume or pour; end unsuccessful and inspect the blender with recording off.

Check: the blender ran for 30 seconds plus 10 seconds only when required. It is off, no fruit piece is visible, and motion has stopped. If either cycle stopped early without an abnormal condition, do not restart, open, or pour; tag the violation and end after motion stops.

Step 6: Open the lid after blending

Goal: open the hinged lid only after the blades stop.

- Confirm no blade or smoothie motion is visible.

- With the **right gripper**, press and release **LID RELEASE**, then move clear.

Check: the lid stays fully open and the pitcher remains seated. If it does not open, press **LID RELEASE** once more with the **right gripper**. Do not remove the pitcher under a closed or partly open lid.

Step 7: Remove and set down the pitcher

Goal: the open pitcher rests upright on the marked table set-down spot.

7.1 Remove the pitcher from the base

- With the **right gripper**, grasp the handle. With the **left gripper**, support the lower wall opposite the handle and away from the spout.
- Lift the pitcher vertically clear of the base. Do not twist or relock it.

7.2 Set down the pitcher

- Carry the pitcher upright to the center set-down spot and lower it until its base rests flat.
- Release the **left gripper**, then the **right gripper**.

Check: the pitcher is upright, stable, and clear of the blender base.

Step 8: Fill the left serving cup in its position

Goal: fill the left serving cup to 1–2 cm below the rim.

8.1 Lift and position the pitcher

- With the **right gripper** on the handle and the **left gripper** supporting the lower wall, lift the pitcher from the set-down spot and center the spout over the left serving cup.

8.2 Pour and stop at the fill range

- Rotate both grippers together and pour a slow, controlled stream. Stop at 1–2 cm below the rim, then return the pitcher upright.

Check: the cup is upright and filled 1–2 cm below the rim. If low, add more. If overfilled, do not pour back; continue to Step 9 and record the cup-fill violation.

Step 9: Fill the right serving cup in its position

Goal: fill the right serving cup to 1–2 cm below its rim and leave both cups equally filled.

9.1 Position the pitcher

- Carry the upright pitcher from the left cup to the right cup and center the spout over its opening. Keep the **right gripper** on the handle and the **left gripper** on the lower wall.

9.2 Pour and finish the batch

- Rotate both grippers together and pour a slow, controlled stream. Stop at 1–2 cm below the rim, then return the pitcher upright. Do not shake or strike it.

Check: both cups are upright, filled 1–2 cm below their rims, and visibly even. Do not transfer smoothie between cups. Leave any excess in the pitcher and record the cup-fill violation.

Step 10: Set down the pitcher, clean spills, and end the episode

Goal: set the pitcher on the table without relocking it, clean spills, and home both arms.

10.1 Set down the pitcher

- Carry the upright pitcher to the marked table set-down spot with the **right gripper** on the handle and the **left gripper** supporting the lower wall. Lower it until the base rests flat.
- Release the **left gripper**, then the **right gripper**.
- Do not return or relock the pitcher on the blender base.

10.2 Clean spills

- Confirm the pitcher is stable on the table and the blender is off.
- If no spill is visible, continue to Step 10.3.
- With the **right gripper**, grasp a cloth corner and wipe spills inward, away from the controls and serving cups.
- Push solids into the waste tray and continue until the table, blender exterior, and cup bases are clear. Never return spilled material to the pitcher or cups.
- With the **right gripper**, return the cloth to the cleanup spot.
- If liquid reached the base or controls, do not touch it. Clear both arms, end, then unplug and clean with recording off.

10.3 End the episode

- Confirm both filled cups are upright in their original right-side positions and no visible spill remains.
- Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

These actions happen while recording is stopped.

- Remove the two filled serving cups and the two empty ingredient cups.
- Unplug the blender. Wash the pitcher and lid under the station procedure, keeping the blade assembly away from the grippers.
- Dry the pitcher, lid, base exterior, table, and cup areas completely.
- Seat the empty pitcher on the base with its handle right and push the hinged lid closed.

- Stage a fresh fruit cup and milk cup in the input zone for the next episode's config — the left (Config L), the front-center (Config M), or the back-right (Config R) — and two clean serving cups in their right-side pouring positions. Clear the pitcher set-down spot.
- Empty the waste tray and replace the cloth if used. Reconnect the blender, confirm it is off, and run both Setup checklists before the next recording.

SOP violations

Actions that break the SOP and are scored in side-by-side review.

How to record a violation in review

For each violation spotted in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken**.

The visible cue is seen in the video. The coaching note supports retraining and is not a label.

Episode handling

Tag each violation with its timestamp and name. Retain the tagged episode in training data.

Violations

Note on the start position: the violations below were written for Config L (ingredient cups at the left, the left gripper loading and the right gripper holding the handle). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the ingredient cups are not in the input zone for the config; a gripper reaches across the table for an ingredient cup or the pitcher handle; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the loading gripper is the one on the ingredient cups' side and the other gripper steadies the pitcher; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the ingredient cups are before the first reach, then follow that config's IF lines through Steps 2 and 3.

Violation: Lid not opened before loading

- **Visible cue:** fruit or milk is moved over the pitcher while the lid is closed or partly open.

- **SOP rule broken:** Step 1 (press LID RELEASE and confirm the lid stays fully open before loading).
- **Coaching note:** open and check the lid before picking up fruit.

Violation: Ingredients loaded in the wrong order

- **Visible cue:** milk enters the pitcher before the fruit.
- **SOP rule broken:** Steps 2–3 (load fruit first and milk second).
- **Coaching note:** follow the fixed left-to-right loading order.

Violation: Ingredient cup not emptied or returned

- **Visible cue:** fruit or milk remains in its cup, a retry is skipped, or an empty cup is not returned to its starting position.
- **SOP rule broken:** Steps 2–3 (empty each ingredient cup, retry when material remains, and return it upright to its starting position).
- **Coaching note:** inspect, retry if needed, and return each cup before continuing.

Violation: Gripper or object enters the pitcher

- **Visible cue:** a gripper, cloth, or other object passes below the rim; or the cup enters farther than needed to guide a rim piece.
- **SOP rule broken:** Steps 2–3 (load by pouring and keep grippers and objects outside the pitcher).
- **Coaching note:** keep grippers and tools above the rim.

Violation: Lid not pushed fully closed before blending

- **Visible cue:** the BLENDING BUTTON is pressed while the lid is open or raised.
- **SOP rule broken:** Step 4 (push the hinged lid straight down and confirm it rests fully closed before blending).
- **Coaching note:** push the lid fully down before using the blending button.

Violation: Gripper inside pitcher footprint at startup

- **Visible cue:** either gripper remains above the lid or pitcher when the BLENDING BUTTON is pressed.
- **SOP rule broken:** Step 5.1 (move both grippers outside the pitcher's footprint before pressing the BLENDING BUTTON).
- **Coaching note:** clear both arms before pressing the blending button.

Violation: Incorrect blend cycle

- **Visible cue:** the initial cycle runs for less or more than 30 seconds; a 10-second extension is unnecessary or skipped; the extension is not 10 seconds; a third cycle starts; or an

interrupted cycle restarts.

- **SOP rule broken:** Step 5 (run 30 seconds, add one 10-second extension only when visible fruit remains, and never run a third cycle or restart an interrupted cycle).
- **Coaching note:** run 30 seconds, inspect at rest, and add 10 seconds only when required.

Violation: Incomplete smoothie poured

- **Visible cue:** the lid is opened or smoothie is poured after the extension while fruit pieces remain visible.
- **SOP rule broken:** Step 5.3 (after the extension, end without opening or pouring if visible fruit remains).
- **Coaching note:** stop if fruit remains visible after the extension.

Violation: Blender touched while blades are moving

- **Visible cue:** a gripper touches the lid, pitcher, or base while motion is visible.
- **SOP rule broken:** Steps 5–6 (keep both arms clear while blending and wait for motion to stop before opening the lid).
- **Coaching note:** keep both arms clear until all motion stops.

Violation: Abnormal blend condition not handled correctly

- **Visible cue:** blending continues after the lid lifts, a leak appears, the pitcher shifts or base moves, or a hard object repeatedly strikes; or blending or pouring resumes afterward.
- **SOP rule broken:** Step 5 abnormal-condition branch (stop immediately, wait for all motion to stop, do not resume or pour, and end the episode unsuccessful).
- **Coaching note:** stop at the first abnormal cue and clear both arms.

Violation: Lid opened incorrectly after blending

- **Visible cue:** LID RELEASE is pressed while contents move, the lid is forced by hand, or the pitcher is removed before the lid is fully open.
- **SOP rule broken:** Step 6 (wait for motion to stop, press LID RELEASE, and confirm the lid stays open before removing the pitcher).
- **Coaching note:** wait, use the lid button, and verify full opening.

Violation: Spill handled incorrectly

- **Visible cue:** spilled material is returned to food, the wet blender base or controls are touched, or a spill remains at episode end.
- **SOP rule broken:** Steps 10.2 and 10.3, discard spills, avoid wet electrical areas, clean after pouring, and confirm the workspace is clear before ending.
- **Coaching note:** clean after setting down the pitcher and never return spilled food.

Violation: Pitcher removed or set down incorrectly

- **Visible cue:** the pitcher is removed with one gripper, twisted, carried tilted, or left unstable away from the marked table spot.
- **SOP rule broken:** Step 7 (use the handle and lower-wall support, lift vertically, carry upright, and set the pitcher flat on the table).
- **Coaching note:** use both grippers and set down the pitcher before pouring.

Violation: Wrong serving-cup order

- **Visible cue:** the first pour goes into the right cup or the second pour goes into the left cup.
- **SOP rule broken:** Steps 8–9 (fill the left serving cup first and the right serving cup second while both remain in their right-side positions).
- **Coaching note:** always work left to right.

Violation: Uncontrolled pour

- **Visible cue:** the spout is outside the cup opening, the stream misses the cup, or the pitcher shakes or strikes another object.
- **SOP rule broken:** Steps 8–9 (center the spout over each cup and use a slow, controlled pour with both grippers).
- **Coaching note:** keep the spout low and centered; rotate both grippers together.

Violation: Serving cup overfilled, underfilled, or uneven

- **Visible cue:** either cup is outside the 1–2 cm fill range, the fills are visibly uneven, or smoothie is transferred between cups.
- **SOP rule broken:** Steps 8–9 (fill both cups to 1–2 cm below the rim, keep them visibly even, and do not transfer smoothie between cups).
- **Coaching note:** watch each surface and stop inside the 1–2 cm range.

Violation: Pitcher relocked or not returned to the table

- **Visible cue:** after pouring, the pitcher is returned or relocked on the blender base, or left unstable away from its marked table spot.
- **SOP rule broken:** Step 10.1 (set the pitcher flat on the marked table spot and do not return or relock it on the blender base).
- **Coaching note:** finish with the pitcher stable on the table.

Violation: Step skipped

- **Visible cue:** a step in the sequence is never performed — only one ingredient is loaded, the blend cycle never starts, the pitcher is poured straight from the base without the Step 7 set-down, a serving cup is left empty, or the pitcher is left on the table with a visible spill uncleaned.

- **SOP rule broken:** Steps 1 to 10 (perform every step in order and finish it, including its **Check**, before starting the next).
- **Coaching note:** name the next step before you move; a step whose check you never made has not been done.

Violation: Wrong arm used

- **Visible cue:** a motion assigned to one gripper is performed by the other, or the prescribed two-gripper lid or pitcher hold is replaced with a one-gripper motion.
- **SOP rule broken:** Steps 1 to 10, each action uses the gripper its step names.
- **Coaching note:** rehearse the left/right roles before recording.

Violation: Wrong episode ending

- **Visible cue:** recording ends before the pitcher is stable on the table, both cups are upright in their original right-side positions, spills are cleared, or both arms are home.
- **SOP rule broken:** Step 10 (set down the pitcher without relocking it, clean spills, confirm both filled cups remain in position, home both arms, and then end data collection).
- **Coaching note:** finish the return and cleanup, then home both arms.

Non-violation failures

Log these failures as system issues, discard the episode, and do not coach on them.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, or motor error.
- **Defective blender:** control failure, motor failure, damaged blade assembly, cracked pitcher, leaking seal, or lid that cannot seat despite the prescribed retry.
- **Defective material:** cracked serving cup, ingredient contamination, or fruit that remains visible after the 30-second cycle and 10-second extension.

Annotation subtasks (from SOP)

1. Open the hinged lid with LID RELEASE
2. Load the fruit and return its cup
3. Add the milk and return its cup
4. Push the hinged lid fully closed
5. Run the 30-second blend cycle and, if required, one 10-second extension
6. Open the lid with LID RELEASE after the blades stop
7. Remove the pitcher and set it on the table
8. Fill the left serving cup in its right-side position to 1–2 cm below the rim

9. Fill the right serving cup in its right-side position to 1–2 cm below the rim
10. Set the pitcher on the table, clean spills, home both arms, and end the episode